

CLASS ACTIVITY 2

AI Feasibility Assumption Audit

Module 2: Project Initiation, Lifecycle Selection, and Feasibility

AzTech

Web and Mobile Application

GRADE

Submitted by:

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TIME 120 min + revision	TEAM 3-4 students	EVIDENCE 5 required outputs	AI POLICY AI-Integrated
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The challenge: AI can make a project appear feasible by inventing costs, compressing schedules, overlooking operational capacity, or treating security and legal risks as minor. Your team must expose and validate assumptions, revise the project plan, and defend a responsible proceed, conditional proceed, re-scope, defer, or no-go decision.

Purpose and learning outcomes

By completing this activity, you will:

1. Convert the Module 1 problem frame and methodology audit into a defensible business need, value proposition, scope, SMART objectives, governance, and success measures.
2. Distinguish constraints, assumptions, dependencies, estimates, and risks before committing to a lifecycle, project plan, or schedule.
3. Evaluate technical, operational, economic, schedule, security, and legal-and-ethical feasibility using traceable evidence.
4. Audit AI-generated feasibility claims and produce a corrected recommendation that strengthens the progressive Title Proposal portfolio.

AI use policy for this activity

AI-Integrated: AI use is required and is part of the process being evaluated. AI is used to challenge assumptions and surface questions; it is not the project approver. The final feasibility decision must be your team's judgment, supported by verified evidence.

1. Use the same project from Class Activity 1. Enter only synthetic, de-identified, or instructor-provided information. Do not enter names, credentials, confidential organizational data, or personally identifiable information into an AI tool.
2. Complete the independent initiation baseline before using AI. This preserves your original reasoning and makes unsupported changes visible.
3. Keep a prompt-and-output record. Identify the AI tool and date used, and preserve only the relevant excerpts needed to show the audit trail.
4. Verify every material claim against Module 2 resources, prescribed readings, project evidence, or another faculty-approved credible source. Unsupported figures, schedules, and risk ratings remain assumptions.
5. Clearly mark what your team accepted, rejected, or corrected. Submit the AI Use Disclosure and take responsibility for the final recommendation.

Important distinction: Diagnostic assessments, formative quizzes, Summative Assessment 1, the midterm examination, live oral checks, and the actual defense remain AI Use Prohibited. This permission applies only to this Class Activity.

Progression rule: Do not start over. Any change to the problem statement, working title, development approach, scope, or safeguards from Class Activity 1 must be supported by new evidence and recorded as a controlled revision.

1. Carry forward Class Activity 1

Use the same specialization-aligned project or instructor-approved case. Reference the corrected recommendation from Class Activity 1 before adding new Module 2 decisions.

Handoff field	Team record
Specialization	Web and Mobile
Working title and version	University Smart Queue & Appointment System (USQAS) — Feasibility & Initiation Baseline v1.0
Sanitized organization or setting	University Student Service Office
Problem or opportunity and expected value	Problem: The queuing process is disconnected from the main database. This causes very long physical lines, repetitive typing of student information, confused wait times, and high stress for office staff. Expected Value: A mobile app for easy scheduling, zero duplicate data typing through database connection, faster queue movement, and happier students.
Module 1 selected approach and rationale	Hybrid Approach: We combine Prototyping (to test screen designs early with students), Agile (for short development cycles), and Predictive SDLC (for strict database security, integration, and university rules).
Safeguards carried forward	• Security: Encrypt data in transit (TLS 1.3) and at rest (AES-256), use OAuth2 login. • Privacy: Minimal data collection (only Student ID, Name, and Service Type); delete activity logs automatically after 30 days. • Accessibility: Follow WCAG 2.1 Level AA rules, make app fast on weak 3G internet, and support screen readers. • Fairness & Inclusion: Put a tablet kiosk in the office lobby and keep paper tickets for students who do not have smartphones. • Intellectual Property: Check all open-source code libraries before using them. • Accountability: Staff always have manual control to adjust queue numbers when needed.
Unresolved evidence from Module 1	• Official wait time logs and manual data entry error numbers from the office. • Information about legacy student database capabilities and API support. • Student phone types, operating systems, and campus Wi-Fi speed data. • Written guidelines from the university Data Privacy Officer (DPO).
Changes after feedback and reason	We added specific WCAG 2.1 Level AA standards, decided to build API middleware before finishing frontend design to protect the database, and scheduled a technical check of the old database.

Team roles

Role	Primary responsibility
Lead analyst	Maintains the business case, scope, lifecycle rationale, and decision trace.
AI interaction recorder	Uses the approved tool and preserves prompts and relevant output excerpts.
Feasibility evidence verifier	Checks claims, estimates, costs, and project conditions against credible evidence.
Risk and ethics challenger	Tests blockers, mitigations, residual risk, and legal-and-ethical conditions.

Handoff check: The problem is more than a symptom; affected stakeholders and expected value are visible; the Module 1 approach and safeguards are traceable; and unresolved evidence is not presented as fact.

2. Establish your independent initiation baseline

Before using AI: Complete this baseline as a team. It preserves your own project judgment and makes unsupported changes to scope, estimates, or feasibility visible.

Business need and value proposition *Beneficiary; current pain or gap; proposed capability; expected outcome; indicator.*

Beneficiary: University students and Student Service Office staff.

Current Pain or Gap: Students wait for 45+ minutes in long lines and have to write their personal details manually on paper or repeated forms. Staff waste around 30% of their working day re-typing student information into disconnected systems.

Proposed Capability: A simple mobile-friendly web app where students can book slots and see live queue times, an admin dashboard for staff to manage calls, and secure API middleware to connect with the student database.

Expected Outcome: Shorter physical crowds, zero repetitive data entry, clear waiting times, and faster office operations.

Indicator: Average waiting time in minutes, queue drop-out rate, database sync error percentage, and System Usability Scale (SUS) survey score.

Independent initiation baseline - continued

Scope boundary

IN SCOPE	OUT OF SCOPE
- Mobile web app for students to schedule appointments and view queue numbers - Calculation system to estimate real time waiting times - Administrative dashboard for staff to call numbers and override queues - API middleware to verify student ID and read/write queue records - Automated Web Push and SMS notifications when turn is close	- Changing or redesigning to core university student database system - Online payment or fee processing systems - Queuing systems for other offices - Building custom physical queue hardware from scratch - Native iOS App Store or Google Play Store mobile apps

SMART objectives

#	Objective	Evidence or measure	Timeframe
1	Reduce Wait Times: Lower average student waiting time in physical line by 40%.	Service office queue logs (From baseline 45 min down to 27 min or less).	Within 60 days after launch.
2	Remove Repeated Typing: Achieve near zero manual re-typing for valid student accounts.	Staff system logs showing automatic student profile loading rate > 98%.	By the end of Sprint 4.
3	Ensure Accessibility: Meet WCAG 2.1 Level AA standard and load under 2 seconds on 3G internet.	AXE accessibility test score of 100%; Google Lighthouse score > 90.	Before Stage-Gate 2 deployment.
4	High User Satisfaction: Get strong feedback ratings from students and office workers.	Post-service digital survey reaching a System Usability Scale (SUS) score > 80/100.	90 days after launch.

Project conditions

Type	Current statement	Evidence / status	Validation action / owner
Constraint	The system must run on existing campus servers and work smoothly on slow campus Wi-Fi.	Known limitation of campus infrastructure.	Conduct network testing using 3G speed limits and choked Wi-Fi.
Assumption	The legacy student database can connect to external apps using REST API or SQL views.	Unverified; need technical access from IT.	Hold an API discovery meeting with Campus IT staff.
Dependency	Written approval and privacy permission from the University Data Privacy Officer (DPO).	Pending submission of privacy assessment forms.	Submit Privacy Impact Assessment (PIA) document to DPO.

Success measures

#	Indicator	Baseline	Target	Source / owner
1	Average Physical Waiting Time	45 minutes	27 minutes or less	Office Queue Logs / Lead Analyst
2	Duplicate Data Entry Error Rate	35% of all requests	Less than 1%	Staff Workstation Audit / Feasibility Verifier

Preliminary lifecycle and feasibility recommendation *State the selected or tailored lifecycle, initial feasibility judgment, and main uncertainty.*

- **Lifecycle:** Tailored Hybrid Approach (Prototyping for UI design → Agile for sprint development → Predictive SDLC for database integration and security checks).
- **Initial Feasibility Judgment:** Proceed with Conditions.
- **Main Uncertainty:** Whether the legacy student database supports direct API connections without causing system slowdowns during enrollment periods.

3. Generate a structured AI feasibility challenge

Use the prompt framework below. Replace bracketed text with your sanitized project brief and include only evidence that your team is authorized to use.

Prompt framework: You are supporting a systems analyst, but you are not the decision-maker. Review the sanitized project brief [insert brief] and challenge - not promote - its feasibility. Use the stated business need, scope, SMART objectives, selected or tailored lifecycle, high-level plan, and known constraints. For each dimension - technical, operational, economic, schedule, security, and legal-and-ethical - (1) separate known evidence from assumptions and dependencies; (2) identify hidden cost, skill, integration, infrastructure, adoption, accessibility, support, approval, data, privacy, intellectual-property, fairness, safety, and accountability issues; (3) explain what may happen if each assumption is false; (4) propose evidence and a practical validation action; and (5) classify the issue as a blocker, condition, risk, or unknown. Also challenge the WBS, milestones, estimate ranges, resource availability, recurring costs, contingency, and residual risks. Do not invent organizational facts or use false numerical precision. End with a provisional recommendation - Proceed, Proceed with Conditions, Revise/Re-scope, or Defer/No-Go - and five questions the analyst must answer before approval.

Prompt-and-output record

Field	Team record
Tool and model/version, if shown	Copilot GPT 5.5
Date and time	September 28, 2026 at 1:49PM
Prompt modifications	Used the 6 dimensional feasibility prompt framework (Technical, Operational, Economic, Schedule, Security, Legal/Ethical) to challenge our baseline ideas instead of approving everything blindly. Asked AI to find hidden costs, risks, and missing technical facts.
Relevant output location or appendix reference	

Priority questions raised by the AI response

#	Question	Why it is material	Owner and validation action
1	Does the old student database have built-in REST API endpoints, or do we need to code our own middleware from scratch?	If there is no API, our coding workload increases by 30-40%, which will delay our 12-week target schedule and increase costs.	Lead Analyst: Conduct a technical meeting with Campus IT database engineers in Week 2.
2	What is the recurring budget cost for SMS message credits during peak enrollment weeks?	Sending thousands of SMS alerts via third-party providers (like Twilio) can waste a lot of money if we do not budget for it.	Feasibility Verifier: Calculate notification numbers and test free Web Push notifications as the primary option in Week 3.
3	How will the system help students who do not have smartphones, active mobile data, or high digital skills?	If we only build a mobile app, poorer students might be left out, which breaks university fairness policies.	Ethics Challenger: Design an on-site lobby kiosk setup and keep physical walk-in tickets ready by Week 3.

Selection rule: Choose the three questions most likely to change scope, cost, schedule, risk exposure, or the decision to proceed.

4. Verify AI assumptions claim by claim

Record at least eight material claims or assumptions, including at least one from each feasibility dimension. A material item is one that could change scope, cost, schedule, risk posture, or the go/no-go decision.

Classification guide: Evidence is supported by a credible source; an assumption is an unverified belief; a dependency relies on another party or resource; and a risk is an uncertain cause-event-impact condition.

#	AI claim or assumption	Classification	Source and validation result	Verdict and corrected feasibility implication
1	The old student database can easily connect through REST API in just 2 weeks.	Assumption		Accept / Reject / Correct Corrected implication:
2	Service office staff will automatically know how to use the admin dashboard without training.	Assumption		Accept / Reject / Correct Corrected implication:
3	Cloud hosting and notification costs will stay under \$50 per month even with high traffic.	Risk		Accept / Reject / Correct Corrected implication:
4	Four 2-week Agile sprints are enough time to complete UI design, database connection, testing, and deployment.	Assumption		Accept / Reject / Correct Corrected implication:
5	Standard HTTPS website security is enough to satisfy student data privacy rules.	Assumption		Accept / Reject / Correct Corrected implication:
6	Collecting student phone numbers for text alerts does not need explicit privacy consent.	Risk		Accept / Reject / Correct Corrected implication:
7	Making a mobile-responsive web app automatically guarantees WCAG 2.1 AA accessibility.	Risk		Accept / Reject / Correct Corrected implication:
8	The new queue system will directly save 40% in labor costs for the university.	Assumption		Accept / Reject / Correct Corrected implication:

Verification standard: A citation alone is not sufficient. Explain how the evidence supports or contradicts the claim. No unsupported cost, schedule, percentage, performance figure, or risk rating may be treated as a fact.

5. Conduct the six-dimensional feasibility audit

Use the validated evidence to reach a separate conclusion for each dimension. Record uncertainty instead of forcing a confident rating.

Dimension	Evidence-supported conclusion	Key limitation or blocker	Required mitigation or condition	Rating and confidence
Technical	Skills, platforms, integration, infrastructure, performance, reliability, supportability	Legacy student database has no ready REST API endpoints.	Build an isolated API middleware layer with read-only database access.	Conditional
Operational	Workflow fit, readiness, adoption, accessibility, ownership, support, change impact	Staff might be afraid of new software or system freezes.	Hold 2 practical training workshops for staff; keep lobby ticket option.	Feasible
Economic	Initial, recurring, indirect, transition, support, retirement, and risk costs; tangible and intangible benefits	Unexpected SMS fees during peak enrollment busy periods.	Set Web Push as default alert; set cloud monthly spending limit caps.	Feasible
Schedule	Effort ranges, resources, dependencies, approvals, milestones, unavailable periods, contingency	Approvals from Data Privacy Officer (DPO) and IT security take long time.	Start DPO review early during Sprint 2 development work.	Conditional
Security	Assets, trust boundaries, threats, safeguards, monitoring, recovery, residual risk	Risk of hackers trying to manipulate queue numbers or steal data.	Use OAuth2 tokens, request limits, and data masking for student details.	Conditional
Legal and ethical	Lawful purpose, consent, privacy, minimization, IP, fairness, safety, and accountable AI	Risk of excluding students who do not own smartphones or data plans.	Place a touchscreen lobby tablet kiosk and keep walk-in support.	Feasible

Most serious cross-dimensional blocker or uncertainty and why

Database API Connection vs. Schedule & Security Bottleneck:

The old student database does not have modern API connections. Rushing this integration without proper security checks could leak student records or break the main database during busy enrollment weeks. But waiting for a full database upgrade from Campus IT will take too many months.

Solution: We will only proceed if we build a safe, read-only API middleware layer with encrypted connection during Phase 1.

Blocker rule: Do not average away a critical legal, ethical, security, or operational blocker. Re-scope, defer, or stop when a required safeguard cannot be lawfully, ethically, or operationally implemented.

6. Stress-test the project plan and risk posture

Revise planning claims that depend on weak assumptions. Use ranges, owners, dependencies, review gates, and contingency rather than wishful dates or unsupported cost figures.

Planning element	Current project claim	AI challenge or hidden assumption	Evidence-based correction	Owner / trigger
Scope and deliverables	Deliver mobile web app, admin panel, SMS engine, and database sync.	Assumes writing data back to old DB is easy; could cause database locks.	Limit Phase 1 DB sync to read-only student lookup; store queue numbers in app DB.	Lead Analyst Trigger: API Discovery Review
Lifecycle and governance	8-week pure Agile sprint plan.	Ignores slow school administration reviews, security checks, and privacy sign-off.	Change to Hybrid model: 3 formal stage gates wrapping 4 Agile sprints.	Lead Analyst Trigger: Charter Approval
WBS and milestones	Launch final app directly at the end of Sprint 4.	Assumes zero software bugs, fast staff learning, and instant sign-off.	Add a 2-week Pilot Phase (Stage-Gate 3) with dual paper/digital queuing before full launch.	Feasibility Verifier Trigger: Sprint 4 End
Estimates and resource availability	2 student coders working part-time for 8 weeks.	Student coders do not have deep experience with security hardening or old databases.	Get a Senior Campus IT staff mentor (5 hours/week) to guide backend middleware setup.	Lead Analyst Trigger: Resource Phase
Dependencies and approvals	DPO privacy approval takes only 3 business days.	School privacy boards usually take 2 to 4 weeks to review software.	Submit Privacy Impact Assessment (PIA) papers right away in Week 1.	Ethics Challenger Trigger: PIA Submission
Cost, total cost of ownership, and expected value	Total project costs under \$500 using free cloud tiers.	Text message fees and extra cloud bandwidth will add monthly expenses.	Limit cloud budget to \$50/month; use free Web Push as main alert, SMS as optional choice.	Feasibility Verifier Trigger: Monthly Audit
Contingency and change control	5% extra schedule buffer (2 days).	Buffer is way too small for school software projects.	Create a 20% schedule buffer (2.5 extra weeks) and formal Change Control Board.	Lead Analyst Trigger: Schedule Freeze

Critical risk register

#	Cause-event-impact risk	Probability / impact rationale	Response and owner	Residual risk / decision trigger
1	Cause: Legacy database has no built-in REST API. Event: App cannot read student data automatically. Impact: Project delayed by 3+ weeks; manual data typing remains.	Prob: High Impact: High Rationale: Old campus systems rarely have modern API setups ready.	Mitigate: Build an isolated Node.js middleware layer using read-only database access. (Owner: Lead Analyst)	Residual Risk: Low. Trigger: Legacy DB connection test fails during Week 2.

#	Cause-event-impact risk	Probability / impact rationale	Response and owner	Residual risk / decision trigger
2	Cause: Too many students log in during enrollment week. Event: App server crashes or runs extremely slowly. Impact: Students get frustrated and lines become long again.	Prob: Medium Impact: High Rationale: Enrollment weeks bring 10 times more web traffic than normal days.	Mitigate: Build simple lightweight code, use Redis caching, and set cloud auto-scale limits. Owner: Feasibility Verifier	Residual Risk: Medium. Trigger: App load speed goes above 2 seconds during stress tests.
3	Cause: Some students do not have smartphones or data. Event: They cannot join the digital queue from outside. Impact: Unfair treatment and complaints from students.	Prob: Medium Impact: High Rationale: Equal service access is required by university rules.	Mitigate: Put an easy lobby tablet kiosk in the office and keep paper tickets available. Owner: Ethics Challenger	Residual Risk: Low. Trigger: Over 5% of students report access problems in pilot tests.

Risk standard: A response must change exposure or readiness. State the response, owner, trigger, action, residual risk, and the person authorized to accept that residual risk.

7. Produce the corrected feasibility recommendation

The final recommendation must integrate the project evidence and your audit. It may agree with AI, but it must not simply repeat the AI conclusion.

Revised working title and version

University Smart Queue & Appointment System (USQAS) — Feasibility & Implementation Plan v2.0

Revised business need, value proposition, scope boundary, and SMART objectives

Lifecycle, WBS, milestone, estimate, and risk changes

Overall feasibility decision

☐ Proceed	☐ Proceed with conditions	☐ Revise / Re-scope	☐ Defer / No-Go
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Decision rationale, confidence, and most important trade-off

Mandatory conditions, mitigations, owners, dates, and decision triggers

Next-gate evidence, decision-change triggers, and how the final recommendation differs from the independent baseline and AI advice

Required outputs

- [] Module 1 handoff and independent initiation baseline
- [] AI prompt-and-output record with three priority questions
- [] Assumption validation record with at least eight material items, including all six feasibility dimensions
- [] Completed six-dimensional feasibility, planning, and risk audit
- [] Corrected feasibility recommendation and signed AI Use Disclosure

Canvas submission: Submit one consolidated PDF named M2_CA2_Section_GroupName.pdf unless your instructor provides another naming or submission rule. Keep the complete AI transcript available for verification; include only relevant excerpts in the submitted evidence pack.

AI Use Disclosure

Disclosure item	Team response
AI tool(s) used	
Purpose of use	Challenge feasibility assumptions; surface hidden costs, dependencies, and risks; generate validation questions; stress-test planning; other: _____
What the AI contributed	
What the team independently contributed	
Claims or assumptions accepted and why	
Claims or assumptions rejected or corrected and why	
Sources and project evidence used for verification	
Decision changes caused by validated evidence	
Data-protection confirmation	We confirm that only synthetic, de-identified, or instructor-provided information was entered.

Team attestation: We reviewed the AI output, validated material claims, and take responsibility for the submitted feasibility analysis, revisions, risk posture, and recommendation.

Name	Signature	Date	Initials
Member 1 / signature		Date	
Member 2 / signature		Date	
Member 3 / signature		Date	
Member 4 / signature		Date	

Evaluation rubric

Criterion	Points	Evidence of mastery
Progressive project initiation and traceability	15	Carries forward Class Activity 1; presents a coherent need, value, scope, objectives, and success measures; all changes are versioned and supported by evidence.
Lifecycle, planning, and risk discipline	15	Tailors the lifecycle; connects WBS, milestones, dependencies, estimates, governance, cost/TCO, risks, owners, and triggers.
Six-dimensional feasibility analysis	25	Evaluates technical, operational, economic, schedule, security, and legal-and-ethical feasibility; identifies blockers, conditions, mitigations, and confidence.

Faculty judgment: A visually complete form does not demonstrate mastery when claims are unsupported, risks are averaged away, or the recommendation cannot be defended from evidence.

Criterion	Points	Evidence of mastery
Assumption validation and evidence quality	20	Validates at least eight material claims across all six dimensions; uses credible sources; explains verdicts; avoids invented facts and false precision.
Professional judgment and corrected recommendation	15	States a defensible decision, conditions, residual risks, next-gate evidence, and change triggers; clearly distinguishes team judgment from AI advice.
AI process evidence and disclosure	10	Prompt record, relevant output excerpts, accepted/rejected/corrected decisions, independent contribution, and disclosure are complete.

Total: ____ / 100

Before you submit

- [] The same project is carried forward, and every major revision from Class Activity 1 is traceable.
- [] Every material AI claim or assumption has a classification, evidence, verdict, and corrected implication.
- [] All six feasibility dimensions are addressed, and no critical blocker is averaged away.
- [] Costs, schedules, performance figures, and risk ratings use evidence, ranges, and stated assumptions rather than false precision.
- [] Conditions and risk responses identify an owner, date or trigger, residual risk, and decision authority.
- [] No confidential, personal, identifying, or credential information appears in the prompt or output record.
- [] The AI Use Disclosure accurately explains the team's independent contribution and final professional judgment.

Alignment source: IT0037 Systems Analysis and Design Syllabus, Module 2 - Project Initiation, Lifecycle Selection, and Feasibility.
 Progressive handoff: Class Activity 1 - AI Methodology and Ethics Audit.